

A proof of Conjecture 00000004780

Swapping the diagonal blocks keeps every joint moment but changes the $M_2(\mathbb{C})$ -valued distribution

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1 The conjecture

Conjecture 00000004780 reads (English version; the Chinese version says the same, with “all joint moments asymptotically identical”):

Definition: Joint moments and operator-valued distributions are two layers of noncommutative invariants. *Conjecture:* There exist two matrix models whose joint moments have identical asymptotics while their operator-valued distributions differ, and the separation is realized by an explicit rearrangement of the block structure.

It is an existence statement. We give two explicit matrix models, the second obtained from the first by swapping its two diagonal blocks, whose joint moments coincide for every matrix size and every noncommutative polynomial, while their $M_2(\mathbb{C})$ -valued distributions differ for every matrix size and in the limit. Everything below is proved in Lean 4 with Mathlib.

2 Definitions

Throughout, N is the size parameter and matrices are written in block form

$$M_2(M_N(\mathbb{C})) = M_{2N}(\mathbb{C}), \quad \text{indexed by } \{0, 1\} \times \{0, \dots, N-1\}.$$

In Lean this is `BMat N := Matrix (Fin 2 × Fin N) (Fin 2 × Fin N) ℂ`. The definitions below are stated for $N \geq 1$. Lean also allows $N = 0$ as a formal extension (the matrices are empty, $b \mapsto b \otimes 1_0$ is not injective and $E_0 = 0$); this level plays no role: every separating statement assumes $N \geq 1$, and it does not affect any limit.

Definition 1 (Matrix model, joint moments). A matrix model of r matrices is a sequence $X = (X_N)_N$ of r -tuples $X_N = (X_{N,1}, \dots, X_{N,r})$ in $M_2(M_N(\mathbb{C}))$ (Lean: `Model r`). Its joint moments are

$$\varphi_N(p) = \text{tr}_{2N}(p(X_{N,1}, \dots, X_{N,r})), \quad \text{tr}_{2N} = \frac{1}{2N} \text{Tr},$$

for every noncommutative polynomial $p \in \mathbb{C}\langle x_1, \dots, x_r \rangle$ (Lean: `jointMoment X N p`, with p in `FreeAlgebra ℂ (Fin r)` and evaluation by `FreeAlgebra.lift`). Taking p a monomial gives the moments $\text{tr}(X_{i_1} \cdots X_{i_m})$; all matrices used below are self-adjoint, so $*$ -moments are moments.

Definition 2 (Operator-valued distribution over $B = M_2(\mathbb{C})$). Let $B = M_2(\mathbb{C})$ act on $M_2(M_N(\mathbb{C}))$ by $b \mapsto b \otimes 1_N$ (Lean: `emb`), and let

$$E_N = \text{id}_2 \otimes \text{tr}_N : M_2(M_N(\mathbb{C})) \rightarrow M_2(\mathbb{C}), \quad E_N(M)_{ij} = \frac{1}{N} \sum_a M_{(i,a),(j,a)}$$

be the block-wise normalized partial trace (Lean: `condExp`). For $N \geq 1$ it is a unital, B -bimodular projection onto $B \otimes 1_N$, i.e. the usual B -valued conditional expectation for 2×2 block matrices. The B -valued distribution of X_N is the family of B -valued moments

$$E_N(b_0 X_{N,i_1} b_1 X_{N,i_2} \cdots X_{N,i_m} b_m), \quad b_k \in B$$

(Lean: `bMoment X N b0 1` with $l = [(i_1, b_1), \dots, (i_m, b_m)]$).

Definition 3 (Rearrangement of the block structure). For a permutation σ of the two block indices, let $P_\sigma = \sigma \otimes 1_N$ be the permutation of $\{0, 1\} \times \{0, \dots, N-1\}$ (Lean: `blockPerm`). The rearranged model is $(\text{rearrange } \sigma X)_{N,i} = P_\sigma X_{N,i} P_\sigma^{-1}$, which moves block (k, l) of $X_{N,i}$ to position $(\sigma k, \sigma l)$ (Lean: `rearrange`, via `Matrix.reindex`).

3 General facts about rearrangements

Lemma 4 (`rearrange_jointMoment`). For every model X , every σ , every N and every p : $\varphi_N^{\text{rearrange } \sigma X}(p) = \varphi_N^X(p)$.

Proof. $M \mapsto P_\sigma M P_\sigma^{-1}$ is an algebra automorphism of $M_{2N}(\mathbb{C})$ (`Matrix.reindexAlgEquiv`), so by the universal property of the free algebra $p(P_\sigma X P_\sigma^{-1}) = P_\sigma p(X) P_\sigma^{-1}$ (`eval_reindex`), and the trace is invariant under reindexing (`trace_reindex_self`). $\square$

Lemma 5 (`rearrange_condExp`, `rearrange_bMoment_first`). $E_N(P_\sigma M P_\sigma^{-1}) = \sigma E_N(M) \sigma^{-1}$ for every M . Consequently, for every model, $E_N((\text{rearrange } \sigma X)_{N,i}) = \sigma E_N(X_{N,i}) \sigma^{-1}$.

So a rearrangement never changes joint moments. On the full B -valued distribution it acts by transport of structure: with $\alpha(b) = \sigma b \sigma^{-1}$, the moment of the rearranged model with coefficients $b_0, \dots, b_m$ equals α applied to the moment of X with coefficients $\alpha^{-1}(b_0), \dots, \alpha^{-1}(b_m)$ (conjugating the output alone, with the coefficients fixed, would be wrong in general). Lean proves only the special case with identity coefficients (Lemma 2), which is all the example below needs. The explicit pair shows that the action can be nontrivial.

4 The explicit pair

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X_N = (A \otimes 1_N, C \otimes 1_N), \quad Y = \text{rearrange } (0 \ 1) X.$$

Thus $X_{N,1} = \begin{pmatrix} 1_N & 0 \\ 0 & 0 \end{pmatrix}$, $X_{N,2} = \begin{pmatrix} 0 & 1_N \\ 1_N & 0 \end{pmatrix}$, and Y swaps the two diagonal blocks: $Y_{N,1} = \begin{pmatrix} 0 & 0 \\ 0 & 1_N \end{pmatrix}$, $Y_{N,2} = X_{N,2}$. All four matrices are self-adjoint (`X_isHermitian`, `Y_isHermitian`). Lean indexes the two matrices by `Fin 2`, so $X_{N,1}$ is `X N 0` and the B -valued moment $E_N(X_{N,1})$ is `bMoment X N 1 [(0, 1)]`.

Theorem 6 (`conjecture4780`). 1. $Y = \text{rearrange } (0 \ 1) X$, and all matrices of both models are self-adjoint.

2. For every N and every $p \in \mathbb{C}\langle x_1, x_2 \rangle$: $\varphi_N^X(p) = \varphi_N^Y(p)$. Both sequences converge, to the same limit $\text{tr}_2(p(A, C))$.

3. For every $N \geq 1$: $E_N(X_{N,1}) = \text{diag}(1, 0)$ and $E_N(Y_{N,1}) = \text{diag}(0, 1)$. Hence the B -valued distributions of X_N and Y_N differ for every $N \geq 1$.
4. The limits of these first B -valued moments are $\text{diag}(1, 0) \neq \text{diag}(0, 1)$, so the limiting B -valued distributions differ as well. (Lean proves the convergence of these first B -valued moments; see the remark on full limiting distributions below.)

Proof. (1) holds by definition and because A, C and 1_N are self-adjoint. (2) The equality is Lemma 1. For the limit: $b \mapsto b \otimes 1_N$ is a unital algebra homomorphism (`embHom`), so $p(X_N) = p(A, C) \otimes 1_N$, and $\text{tr}_{2N}(M \otimes 1_N) = \frac{\text{Tr } M \cdot N}{2N} = \text{tr}_2(M)$ for $N \geq 1$ (`ntr_emb`, `jointMoment_X`); so the sequence is eventually constant. (3) The B -valued moment with $b_0 = b_1 = 1$ and the single letter x_1 is $E_N(X_{N,1}) = E_N(A \otimes 1_N) = A$ (`condExp_emb`); by Lemma 2 the same moment of Y is $\sigma A \sigma^{-1} = \text{diag}(0, 1)$. (4) Both sequences are eventually constant. $\square$

5 Remarks on the reading

- *Identical asymptotics.* We prove equality of the joint moments for every N , which gives identical asymptotics under any reading (same limits, same rates, same expansions), and we also prove that the limits exist.
- *Operator-valued distribution.* We use the standard $M_2(\mathbb{C})$ -valued distribution of a 2×2 block matrix model with respect to $E_N = \text{id}_2 \otimes \text{tr}_N$. The two models already differ at the first B -valued moment, which belongs to every version of the B -valued distribution (with or without B -coefficients, finite N or limit).
- *Full limiting distributions.* The Lean theorem states convergence only for the first B -valued moments. The full limiting distributions do exist: for $N \geq 1$, multiplicativity of $b \mapsto b \otimes 1_N$ and $E_N(b \otimes 1_N) = b$ give $E_N(b_0 X_{N,i_1} b_1 \cdots X_{N,i_m} b_m) = b_0 g_{i_1} b_1 \cdots g_{i_m} b_m$ independently of N , with $(g_1, g_2) = (A, C)$ for X and $(\sigma A \sigma^{-1}, C)$ for Y . This observation is not formalized; the separation itself only needs the first moment.
- *Rearrangement.* The second model is literally the first with its block indices permuted. Lemma 1 holds for every model and every σ , in particular for every sample of a random matrix model.
- *Degenerate models.* The explicit models are deterministic (constant random matrices) and have the form $a \otimes 1_N$. The conjecture does not require randomness or a particular ensemble.

6 Lean formalization and axiom audit

The file `lean/Conjecture4780/Basic.lean` (248 lines, Lean 4.33.1, Mathlib v4.33.1) contains `ntr`, `Model`, `eval`, `jointMoment`, `condExp`, `emb`, `bMoment`, `blockPerm`, `rearrange`, the lemmas `rearrange_jointMoment`, `rearrange_condExp`, `rearrange_bMoment_first`, `condExp_emb`, `ntr_emb`, `jointMoment_X`, and the main theorem `Conjecture4780.conjecture4780`. It builds with `lake build`. For each of `conjecture4780`, `rearrange_jointMoment`, `rearrange_condExp` and `rearrange_bMoment_first`, `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.